1 A student investigates the motion of a trolley on a wooden surface, as shown in Fig. 1.1.

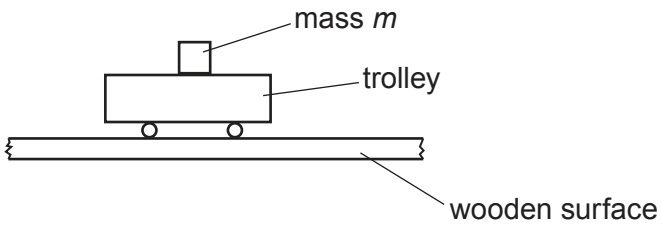

Fig. 1.1

A mass *m* is placed on the trolley.

A mass *P* is attached to the trolley by string which passes over a pulley. When this mass falls, it pulls the trolley along the surface.

The trolley is initially at rest. The student investigates how the speed *v* of the trolley at a distance *d* from the initial position of the trolley varies with *m*.

It is suggested that the relationship between *v* and *m* is

$$\frac{2d}{v^2} = \frac{m + R}{Pg - Q}$$

where *g* is the acceleration of free fall and *Q* and *R* are constants.

Design a laboratory experiment to test the relationship between *m* and *v*.
Explain how your results could be used to determine values for *Q* and *R*.

You should draw a diagram, on page 3, showing the arrangement of your equipment. In your account you should pay particular attention to:

- the procedure to be followed
- the measurements to be taken
- the control of variables
- the analysis of the data
- any safety precautions to be taken.

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